RESEARCH NOTE · 10 OCTOBER 2026

Video Jev: findings so far, and a plan to generalise

A typed, calibrated audio-visual model that answers structured questions about a video in one forward pass.

Repo: multimodal-jev Compute: Modal (H200) Spend to date: $156.75 billed

SUMMARY

- We trained Qwen3-Omni (30B MoE, 3B active) with LoRA and a typed answer head on SnapUGC, a dataset that records how many viewers keep watching short videos.
- On 1,736 held-out test clips, it scores **0.709** (mean of Spearman and Pearson). That is +0.065 over our strongest baseline, a CLIP-based model (95% CI 0.041 to 0.088).
- The best entry in the VQualA 2025 challenge scored 0.710 with an ensemble of four models. Our single model is level with it, but on a different test split, so we cannot yet claim parity.
- Its probabilities are well calibrated (ECE 0.028). That means code can act on how confident it is, not just on the answer.
- The same design can take any question written as yes/no, choice or score. To ask "when" questions (for example "when is the goal?"), we need one new question type. Section 5 is the plan.

1. What the model sees

The model does not take raw mp4 files. Each video is decoded once, and the model reads the frames and audio together on one timeline.

mp4 file
Any length, any codec.

→

Decode (ffmpeg)
First 16 s, 2 frames per second, longest side 448 px, 16 kHz mono audio. Saved once to a cache.

→

Jev record
State (for example title, prompt, duration) plus the questions, written as text in Jev form.

→

Qwen3-Omni
Frames and audio are interleaved by timestamp, with the text after them. LoRA on the text attention.

→

Typed head
From Clef: one logit per allowed answer, for every question. Softmax gives probabilities.

So "raw video" in the useful sense: pixels and sound, with no hand-made features. For SnapUGC the record also carries the title and description. For VideoFeedback it carries the text prompt that made the video.

Question types available today

TYPE	OUTPUT	EXAMPLE	STATUS
`noul`	P(yes), P(no)	"Does this video look AI-generated?" "Did the audience laugh?"	works now
`choice`	Probability per named option	"Which sport is this?" "Which part is funniest: start, middle or end?"	works now
`score`	Probability per ordered level	"How realistic are the hands, 1 to 4?" "What share of viewers keep watching?"	works now
`moment`	Probability per time step	"At what second does the goal go in?" "When is the funniest moment?"	to build

Many questions share one forward pass, so asking ten questions costs about the same as asking one.

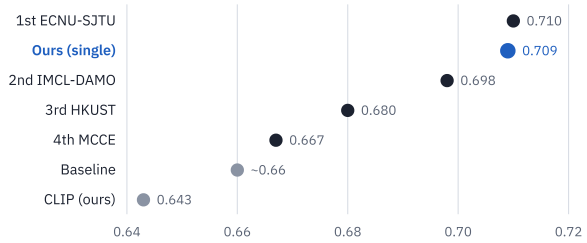
2. Results on SnapUGC

Test split, scored once (1,736 clips)

MODEL	FINAL	BRIER	ECE
Qwen3-Omni + typed head	**0.709**	**0.818**	**0.028**
Omni + CLIP ensemble	0.709	0.816	0.033
CLIP frames + MLP	0.643	0.843	0.034
CLIP + text + MLP	0.639	0.859	0.029
Text only (XGBoost)	0.347	0.884	0.029
Clef-flash, no training	0.127	0.912	0.064

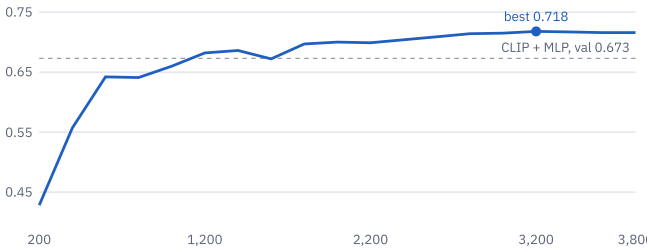
Final = mean of Spearman and Pearson (higher is better). Brier and ECE: lower is better. A uniform guess over 10 levels has Brier 0.900.

Against the VQualA 2025 challenge



Axis starts at 0.64 to show the gaps. Challenge scores are on the official test set; ours are on our own held-out split. The winner ensembled three VideoLLaMA2 models and one Qwen2.5-VL model.

Training curve (full run, 30,631 clips)



Validation final score (500 clips) against training step. Learning rate follows a cosine to zero. Earlier runs at a higher rate collapsed; 3e-5 was stable.

What we learned

- **Training is what matters.** The same family of model with no training scores 0.13. Trained, it scores 0.71.
- **Learning rate decides stability.** A LoRA rate near 1e-4, held for 400+ steps, collapsed twice. 3e-5 trained cleanly for the full epoch.
- **Audio helps a little at most.** Removing audio cost 0.026 at 10k clips, inside the noise. A clean test is still open.
- **Ensembling stopped helping.** Adding CLIP helped weaker Omni models; on test it adds +0.001.
- **Calibration comes for free.** The loss (cross-entropy + Brier + ordinal distance) gives ECE near 0.03 with no extra step.
- **Limits:** one seed only; test split is ours, not the challenge's.

3. Latency, hardware and cost

ITEM	VALUE	SOURCE
Forward pass, one 16 s clip with audio	about 290 ms	Measured, p50, batch 1, H200
Speed against real time (16 s windows)	about 55x	From the line above
Training throughput, video + audio	2,860 clips/h	Measured
Training throughput, video only	5,200 clips/h	Measured
GPU cost on Modal	about $4.80/h	Measured ($17 for 3.55 h)
Inference cost per 1,000 clips	about $0.40	Estimate: 290 ms × 1,000 at $4.80/h, decode excluded
Memory, bf16 weights	about 62 GB	Estimate: needs one 80 GB+ GPU (H100, A100 80 GB, H200)
Memory, 8-bit / 4-bit	~32 / ~17 GB	Estimate, not tested: 48 GB card / 24 GB consumer card
Full training run (30k clips, 1 epoch)	7.5 h, ~$43	Measured

Long videos. Cost grows with length, since every frame adds tokens. A 10 minute video at 2 fps has about 37 times the frames of a 16 s clip. Plan: split long videos into windows (for example 16 to 30 s, with overlap), score each window, then combine. The MoE design keeps per-token compute low (3B active), which is why a 30B model runs in under a third of a second.

What a product needs. One H100 or H200 can serve about 12,000 clips per hour at batch 1, and more with batching. For live use (for example spotting a goal), 16 s windows score well ahead of real time on one GPU.

4. Is this new?

Parts of this exist. The combination is what we have not found.

PRIOR WORK	WHAT IT DOES	WHAT IT LACKS
VideoScore (TIGER-Lab, 2024)	Scores AI videos on 5 fixed dimensions with a regression head.	Fixed questions, no audio, no probabilities.
VQualA 2025 entries	Predict engagement with video-language model ensembles.	One task, one number, no calibration, several models.
Timestamp models (TimeChat, Qwen2.5-VL and others)	Generate timestamps as text.	No probability over time, one answer per generation.
Likelihood scoring of choices	Score several choice answers in one pass.	Text-like answers only, not trained for calibration.
Clef-flash (Cloudflare)	Typed head over images and video for Jev questions.	No audio; on SnapUGC it scores 0.13 untrained.

Our candidate claim: one audio-visual model that answers any question written in Jev form (yes/no, choice, score, and soon moment), many per forward pass, with calibrated probabilities, trained jointly across tasks. The strongest version of the claim: it answers *new questions it was not trained on*, written at test time. A proper literature review is step one, before we state this in a paper.

5. Plan to generalise

Each phase has a small test first, and a gate that decides whether to continue. Costs are GPU estimates at the rates in section 3.

G0 · AI-video quality (VideoFeedback)

`~$6 small · ~$30 full`

5 score questions per clip: visual quality, temporal consistency, motion, prompt match, physics and anatomy. 37.6k clips, Apache-2.0. Already downloaded, indexed and half decoded.

Gate: the small run (2k clips) reaches average Spearman 60 or more on the official test. The full run targets VideoScore's 77.1. GPT-4o scores 23.

G1 · "Does this look AI?" and anatomy

`~$15`

Add `noul` questions for real against generated (GenVidBench and similar) and hand or body faults. Train jointly with G0 in one model.

Gate: joint model keeps G0 scores within 0.02, and beats a CLIP baseline on real-vs-AI accuracy and calibration.

G2 · New question type: `moment`

`~2 days of code · ~$20`

The video tokens already sit in time order, each tied to a timestamp. The head scores each time step against the question and returns a probability over time. That answers "when" in one pass, with confidence, and with the option to report the top few moments.

Data: QVHighlights (per-clip highlight scores and moment spans), Charades-STA (moment spans).

Gate: within 5 points of published moment-retrieval models on QVHighlights (R1@0.5, mAP), using one shared model.

G3 · Sound and humour

~$20

Questions where audio should matter: "did this get a laugh?" (`noul`), "when is the laugh?" (`moment`), "which sound is this?" (`choice`).

Data: SMILE and UR-FUNNY (laughter), FunQA (funny moments with timestamps), VGGSound or AudioSet (sound events). Sport: SoccerNet for goal timestamps (needs a signed agreement).

Gate: removing audio drops these tasks clearly (unlike SnapUGC). This is the clean audio result we lack.

G4 · One joint model, and new questions at test time

~$60 to $100

Train one model on all tasks above plus SnapUGC. Hold out whole question families (for example, train on "is the hand realistic", test on "are the teeth realistic").

Gate: on held-out questions it beats the same model untrained and an off-the-shelf model given the same question. This is the headline result for the paper.

G5 · Long video and speed

~$10

Window long videos and cache the encoder output, so new questions on a seen video cost only the head. Measure latency and cost per video hour.

Rules that carry over

- Official test splits where they exist, scored once per final model. 3 seeds for headline numbers.
- Every task gets a cheap baseline on the same split (CLIP + MLP at minimum).
- Only data with a licence that allows research use. Many sets are built from YouTube, so we check each one and do not redistribute videos. No private data.
- Log every run in `results/README.md` and W&B, with cost.

Risks

- **Tasks may fight.** Joint training can lower single-task scores. Mitigation: balance task sampling, check G1 gate.
- **Licences.** Some sets (SoccerNet, YouTube-based ones) limit use. Mitigation: start with Apache or CC sets.
- **Long-video cost.** Mitigation: windows, lower frame rate for slow questions, encoder cache.
- **Overclaiming novelty.** Mitigation: literature review before writing.

6. Budget and decisions

PHASE	GPU ESTIMATE
G0 small test, then full	$6 to $36
G1 real-vs-AI and anatomy	$15
G2 moment type	$20
G3 sound and humour	$20
G4 joint model + held-out questions	$60 to $100
G5 long video	$10
SnapUGC: 3 seeds + same-split rerun of the challenge winner	$90 to $120
Total	**about $220 to $320**

Decisions needed:

1. Raise the Modal spend limit (about $300 covers the whole plan with care).
2. Order: G0 then G2 (moment type, the most novel part), or G0 then G1 (AI detection, closest to Retina's needs).
3. Whether SnapUGC seeds and the challenge rerun come before or after G0.